

ABHISHEK LALWANI

AI Engineer | Generative AI, LLMs & Agentic AI | Enterprise AI Solutions | RAG & AI Systems Engineering

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AI Engineer with over 4+ years of experience in Generative AI, LLMs, AI application development, and enterprise AI solutions, seeking opportunities in Enterprise Software & Technology. Targeting roles across Generative AI & LLM Engineering, Agentic AI and Front-End/Web Development.

Profile Summary

- Gained expertise in Artificial Intelligence, specializing in Generative AI, enterprise AI integration, production systems, multi-agent orchestration, and semantic search; pursuing M.Tech. in AI/ML from BITS Pilani.
- Currently working as Senior Software Engineer at Dassault Systèmes, leading Generative AI initiatives spanning LLM assistants, enterprise integrations, and cross-functional MLOps practices.
- Delivered a production-grade RAG pipeline that reduced document retrieval time by 60%, improving information accessibility and user productivity.
- Developed LLM chatbot workflows serving 1,000+ users, while building GraphMind, a full-stack knowledge-graph retrieval platform as part of the M.Tech. AI/ML project at BITS Pilani.
- Applied MCP, LangGraph, Hermes, and agent-consumable endpoints to enable seamless AI-agent integration with enterprise applications and multi-agent workflows.
- Exhibited strengths as a Full-Stack AI System Builder & Integrator, with focus on Production-Grade AI Deployment & MLOps and Performance Optimization & Scalability.

Core Competencies

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|---------------------------|--------------------------------|-----------------------------|
| Generative AI Strategy | AI Solution Architecture | Data-Driven Decision-making |
| Intelligent Automation | AI Deployment Management | AI Solution Architecture |
| Agentic Workflow Design | Model Lifecycle Management | AI Product Development |
| Enterprise AI Integration | Cross-Functional Collaboration | Stakeholder Management |

Technical Skills

- AI/ML:** LLMs (GPT-4, Claude, Llama, Mistral), Production RAG Systems, LLM Integration, Agentic AI, LangGraph, Hermes, Multi-Agent Orchestration, Model Context Protocol (MCP), Semantic Search, Embeddings, Model Evaluation, LLM Inference, Knowledge Graphs, LangChain, LlamaIndex, Vector Databases (FAISS, Pinecone), Prompt Engineering, Fine-Tuning, NLP, MLOps, Ollama
- Backend:** Python, FastAPI, Node.js, REST APIs, Full-Stack Development, Celery, Redis, Neo4j (NoSQL), SQL, Data Pipelines, MLflow
- Frontend:** Typescript, JavaScript, Lit Element, Polymer, Redux, Web Components, HTML5, CSS3
- Tools:** Docker, CI/CD, Unit & Integration Testing, Git, Agile, Scrum, scikit-learn, PyTorch, Pandas, NumPy

Work Experience

Dassault Systèmes, Pune | Jul 2022 - Present
Software Engineering Specialist | Jul 2022 – Mar 2025
AI/ML Engineer – Generative AI & Agentic Systems | Apr 2025 - Present

- Designing and deploying enterprise AI applications using Python, FAISS, embeddings, and semantic retrieval, including a production documentation assistant that fetches relevant user documentation and delivers contextual query responses.
- Developing an AI-powered orchestration agent for production failure debugging, enabling automated log analysis, issue identification, and remediation recommendations using relevant documentation.
- Operating LLM-powered assistants using GPT-4, Llama, and Mistral through LangChain-based application workflows, supporting conversational and agentic enterprise use cases.
- Developing and exposing an MCP server for the application, enabling AI agents to interact with application capabilities and supporting agent-driven workflows and tool integrations.
- Orchestrating tool-calling sequences with human-in-the-loop checkpoints across complex AI task flows, enabling controlled execution and validation of agent-driven actions.
- Collaborating with product and domain teams to translate business requirements into scalable AI and Generative AI solutions, integrating intelligent automation into enterprise application workflows.
- Automating end-user query and support workflows through conversational AI, reducing manual effort and delivering measurable productivity gains of 5–10 hours per week.

Software Engineering Specialist | Jul 2022 – Mar 2025

- Owned 6 enterprise applications, modernizing them with the latest technologies and architecture; led major migrations of 2 apps, including a business performance monitoring platform and a no-code UI builder.
- Engineered high-performance web components using Lit Element, Polymer, and TypeScript, enabling scalable front-end experiences across enterprise applications.
- Modernized legacy architecture with Redux and TypeScript, enhancing UI/UX and driving 2–3x growth in customer adoption.
- Delivered GenAI solutions across the migrated applications, including a drag-and-drop GUI platform that enabled users to create and integrate custom applications without coding.

- Optimized rendering pipelines through performance profiling, reducing application load time by 50–75% while strengthening application performance.
- Implemented unit and integration testing frameworks to strengthen regression coverage and release confidence across applications.
- Strengthened customer retention and acquisition by delivering applications that enhanced business monitoring and application-creation capabilities, increasing field-team confidence in positioning the solutions with prospective customers.

Projects

GraphMind - Knowledge Graph & Retrieval Platform | Jun 2026 – Present

- Architected and deployed a full-stack knowledge graph platform using FastAPI, Neo4j, Celery, Redis, and Lit/Redux, transforming documents, code, audio, video, and repositories into query able knowledge graphs.
- Conducted a cross-model evaluation across 3 LLMs and 35 questions, demonstrating improved retrieval faithfulness while reducing token costs ~2.4x through a graph-first retrieval mode.
- Developed an MCP server exposing entity search, graph traversal, and retrieval capabilities as JSON-RPC tools for MCP-compatible clients; added graph versioning, lineage, and deterministic tabular-data extraction with human-in-the-loop review, reducing input tokens ~2x.

Final Year Project - NGO Revenue Forecasting & Donor Segmentation | Jun 2022

- Developed an NGO revenue forecasting and donor segmentation solution using SARIMA, RFM, and K-Means to support data-driven revenue prediction and donor profiling.
- Published the project in the International Journal of Computer Applications (IJCA) in Jun 2022.

Education

- **M.Tech. - Artificial Intelligence & Machine Learning** | BITS Pilani WILP | 2026
- **B.Tech. - Information Technology** | Savitribai Phule Pune University (MMCOE) | 2022

Achievements

- Earned the Applause Award – H1 2026 at Dassault Systèmes for contributions to Performance Study Migration and MCP Tools.
- Received the Applause Annual Award 2024 at Dassault Systèmes as a core team member for Inclusive Simulation Process Automation.
- Secured the Genius Award – H2 2023 at Dassault Systèmes in recognition of individual excellence and impactful contributions.
- Earned Star Alumni recognition from IIT Bombay’s Spoken Tutorial Program for the 2022 batch in 2023.
- Ranked as a Finalist in Smart India Hackathon 2020 for developing an AI-based career recommendation system for school students.

Personal Details

- Languages Known: English, Hindi, Marathi